Experiment 4 Analysis Report: Rutherford Scattering

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The differential cross-section of alphas scattering off a gold target varies over energy and angle, as found by Rutherford. Theoretically, this is proportional to $1/E_0^2 \cdot 1/sin^4(\theta/2)$, and the angular-dependence was verified by this experiment. The data showed a 25% lower systematic shift from the theoretical values, but the trend still matches $1/sin^4(\theta/2)$. The composition of the scattering target was also found by utilizing the separation of the different scattering peaks from the kinematics of scattering off different nuclei. The target's composition was found to be 79.1(1.0)% Gold, 11.60(61)% Silver, and 9.27(86)% Copper.

I. BACKGROUND INTRODUCTION

This experiment uses Rutherford scattering of alpha particles off a mostly Gold target to measure how the experimental cross-section varies with scattering angle and incident beam energy. This is a common method used for material analysis because you can look at the behavior of the scattered alphas to piece together the composition of the scattering target. The experiment was performed at the Edwards Accelerator Laboratory utilizing the 4.5 MV tandem Van de Graaf accelerator and RBS chamber. During the experiment, a movable detector was varied from -135 to 155 degrees from the beam axis within the RBS chamber for a set energy to measure the angular dependence of the differential cross-section. The Energy dependence was tested by changing the incident beam energy between 2.5 to 6.5 MeV.

II. THEORY

The scattering cross-section, σ, is used as a measure of how likely an interaction is to take place between two nuclei. When used for describing nuclear reactions, it is similar to the geometric cross-section in that it relates the number of scattered particles to the number of incident particles and the number of particles that are present to be scattered off of. This probability of interaction is recorded using the unit of barns, where 1 barn = 10^{-28} m^2.

Rutherford scattering uses the Coulomb potential between two nuclei to find how the differential cross-section changes with scattering angle, nuclei charges, and the energy of the system. This energy is typically the average energy of the beam particle after it has traveled through a portion of the scattering foil. As it travels through, it loses some of its energy, and the average energy needs to be properly calculated using the stopping power information from [2] and Eqn. 1. Once the beam particle has scattered, its measured energy depends on the scattering angle via Eqns. 2 and 3.

$$E_{\text{avg}} = \frac{E_i + E_f}{2} \tag{1}$$

$$E_0' = KE_0 \tag{2}$$

$$K = \left(\frac{M_0 cos\theta + \sqrt{M_1^2 - M_0^2 sin^2\theta}}{M_0 + M_1} \right)^2 \tag{3}$$

Once the energetics of the system have been calculated, Eqn. 4 can be used to find the theoretical differential cross-section where Z_0 and Z_1 represent the charges of the beam and target nuclei, respectively. The masses M_0 and M_1 also represent the masses of the beam and target nuclei, respectively.

$$\frac{d\sigma}{d\Omega}\left(\frac{\text{mb}}{\text{sr}}\right) = 1.296\left(\frac{Z_0 Z_1}{E_0}\right)^2 \left[\frac{1}{sin^4(\theta/2)} - 2\left(\frac{M_0}{M_1}\right)^2\right] \tag{4}$$

This theoretical differential cross-section can be compared to the experimental one given in Eqn. 5 where N_s is the total number of scattered nuclei, N_0 is the total number of nuclei impinged on the foil, $\Delta\Omega$ is the detector's solid

angle, and n_1 is the areal density of the target foil.

$$\frac{d\sigma}{d\Omega} = \frac{N_s}{N_0 \cdot \Delta\Omega \cdot n_1} \tag{5}$$

This works great for pure targets of the same nucleus, but for targets composed of different nuclei, the total target differential cross-section can be found by Eqn. 6 where f_i is th fraction of the areal density of the target for a specific nucleus, i [1].

$$\frac{d\sigma}{d\Omega} = \sum_i \left(\frac{d\sigma_i}{d\Omega_i} \cdot f_i \right) \tag{6}$$

III. EXPERIMENTAL METHODS

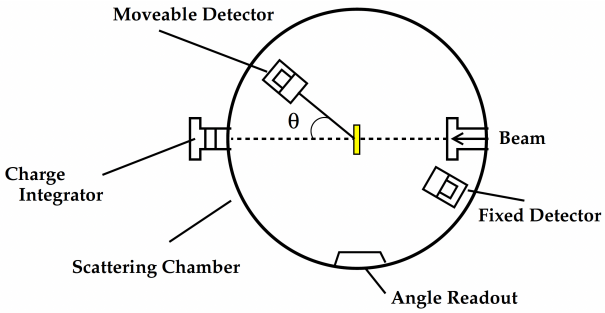


FIG. 1: RBS scattering chamber setup [1].

For this experiment, a beam of 2+ ionized ^{4}He, made at the Edwards Accelerator Laboratory, was sent into the RBS scattering chamber seen in Fig. 1. A target foil of unknown composition was used as the scattering target while a range of beam energies were used to bombard the target foil and measure the scattered alphas at a variety of angles. For every angle measurement from the movable detector, a similar measurement of the fixed detector was recorded, which was positioned at 168 degrees from the beam axis. When the beam energy was 2.5 MeV, a variety of angles were measured spanning -135 to 155 degrees to get the angular distribution of the scattering cross-section. It should be noted that the target was rotated to be half of the detected angle when the movable detector was less than 90 degrees from the beam axis. The energy distribution was also probed by using 50 and 155 degree measurements for beam energies of 3.5, 4.5, and 6.5 MeV.

IV. ANALYSIS AND RESULTS

A. Energy Calibration

We performed an energy calibration by finding the center of the gold peak from the 50 and 155 degree data sets from the 2.5, 3.5, 4.5, and 6.5 Mev beam energy runs shown in Fig. 2. This was done by using Eqns. 1 - 3 and the stopping power information found in [2] to calculate the proper scattered alpha energy to be used for the calibration. E_f was calculated assuming a foil composition that was 80% Gold, 10% Silver, and 10% Copper. The results of which can be seen in Fig. 3 and Tables I - II.

TABLE I: Energy calibration linear fit parameters.

Slope (bins/MeV)	84.52(15)
Intercept (bins)	-8.99(48)

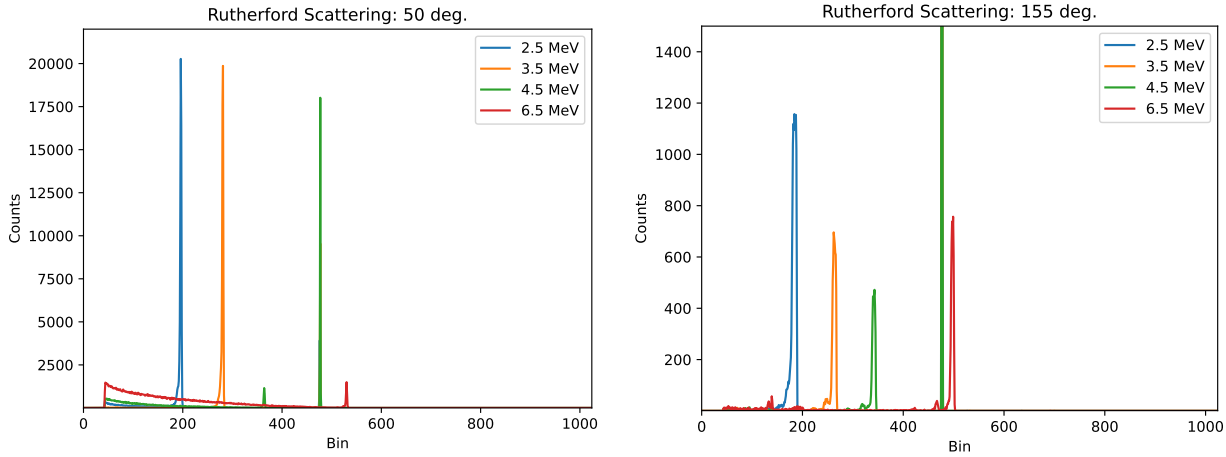


FIG. 2: 50 and 155 degree measurements at different energies.

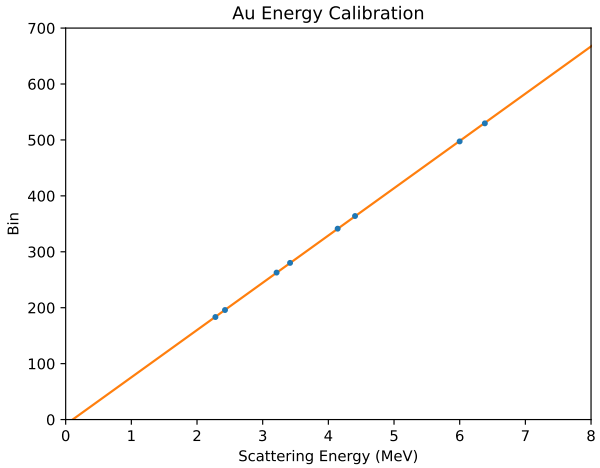


FIG. 3: Energy calibration points and linear fit result.

TABLE II: Energy calibration fit results.

Scattering Energy (MeV)	Bin	Residual (bins)
2.28	183.35	-0.40
2.43	195.86	-0.22
3.21	262.81	0.34
3.42	280.15	0.31
4.14	341.32	0.25
4.41	363.94	0.47
6.00	497.33	-0.75
6.38	529.65	-0.88

B. Target Composition

The composition of the target was experimentally found by using the 155 degree data for an alpha beam energy of 6.5 MeV. This data set was chosen for the clear separation of the peaks from the high beam energy and extreme backward angle. The total number of counts in each peak were found by taking the integral of the histogram over the range of each peak and correcting by the dead-time. The raw data used for this can be seen in Fig. 4 where the Copper, Silver, and Gold peaks can be seen left-to-right. Eqns. 4 and 7 - 10 were used to calculate the foil's composition as seen in Table III.

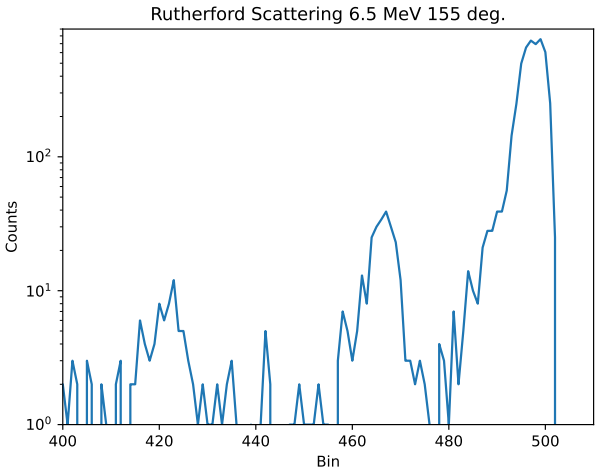


FIG. 4: Raw data for the 155 degree 6.5 MeV alpha beam.

$$f_{Y/X} = \frac{N_Y}{N_X} \frac{(\frac{d\sigma}{d\Omega})_X}{(\frac{d\sigma}{d\Omega})_Y} \tag{7}$$

$$\delta f_{Y/X} = |f_{Y/X}| \cdot \sqrt{\frac{1}{N_X} + \frac{1}{N_Y}} \tag{8}$$

$$f_{\mathrm{X}} = \frac{1}{1 + f_{Y/X} + f_{Z/X}} \tag{9}$$

$$\delta f_X = f_X^2 \cdot \sqrt{\delta f_{Y/X}^2 + \delta f_{Z/X}^2} \tag{10}$$

TABLE III: Foil composition results from 155 degree 6.5 MeV data.

	Counts	Composition %
Gold	4848	79.1(1.0)
Silver	251	11.60(61)
Copper	76	9.27(86)

C. Differential Cross-section Angular Distribution

The angular distribution of the alpha scattering was taken using the 2.5 Mev beam energy, and the raw data can be seen in Fig. 5. For the calculation of each angle's differential cross-section, the stopping power and newly found foil composition were used to calculate the scattered alpha energy for interactions with the Gold, Silver, and Copper in the foil. This energy was used with the energy calibration to find the central bin each peak should show up at as seen in Fig. 6. Using these as a reference, the integral of the histogram was taken over the range of all peaks using bounds that were equally spaced away from the Copper (left) and Gold (right) peak centers. This integral was corrected by the recorded dead-time and was used as N_s in Eqn. 5. The results of this calculation can be seen in Fig. 7(a). The theoretical differential cross-section was calculated using Eqn. 6 with a 1σ error band from the error of the foil's total composition.

The experimentally-found differential cross-section was systematically lower than the theoretical calculation, but there were major discrepancies in the 40, 50, and 60 degree data with a slightly larger than average discrepancy for

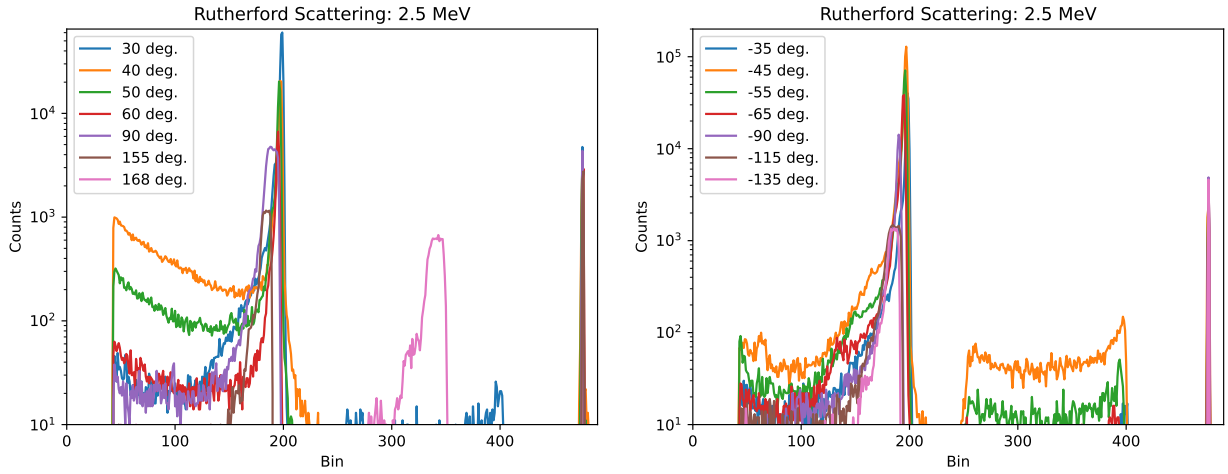


FIG. 5: Raw data for angular distribution measurements with a 2.5 MeV alpha beam.

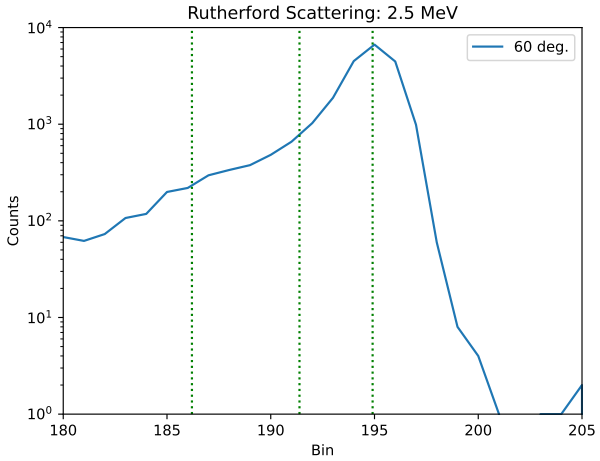
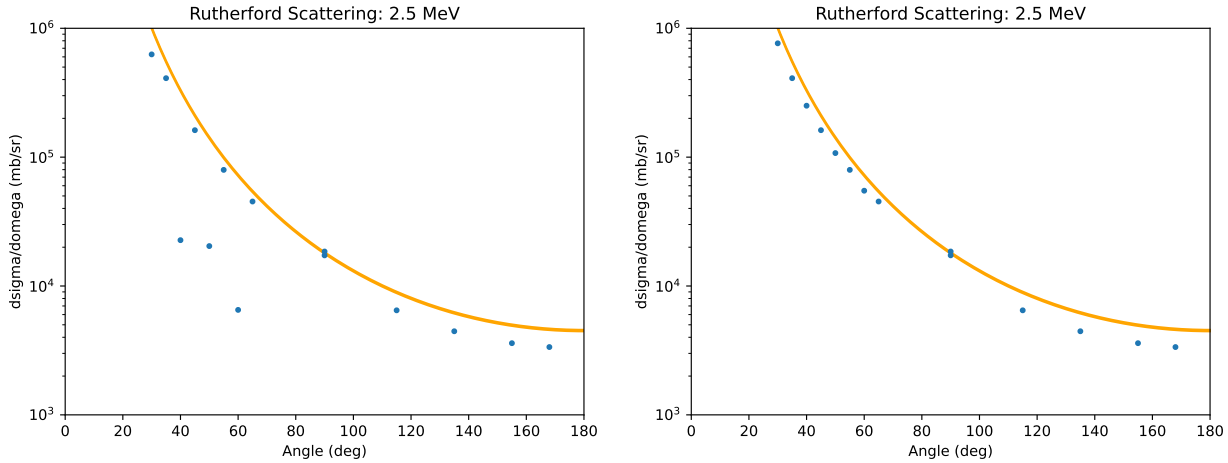


FIG. 6: 60 degree data showing the bin positions where the Cu, Ag, and Au peaks should be left-to-right.

the 30 degree data set. This difference can be seen in Fig. 8(a). It was thought that the scattering rate of alphas was overwhelming the detector, so the dead-time that was recorded was incorrect. In order to be more consistent with the rest of the data, an expected dead-time was calculated to adjust the original measurements. The average of the systematic shifts for the four most consistent lower and four higher angles was used to adjust the final results for the 30, 40, 50, and 60 degree data sets. From these newly estimated results, the expected dead-time was calculated and can be seen in Table IV. The final corrected results can be seen in Figs. 7(b) and 8(b).

TABLE IV: Experimental vs theoretical differential cross-section ratio and dead-time results.

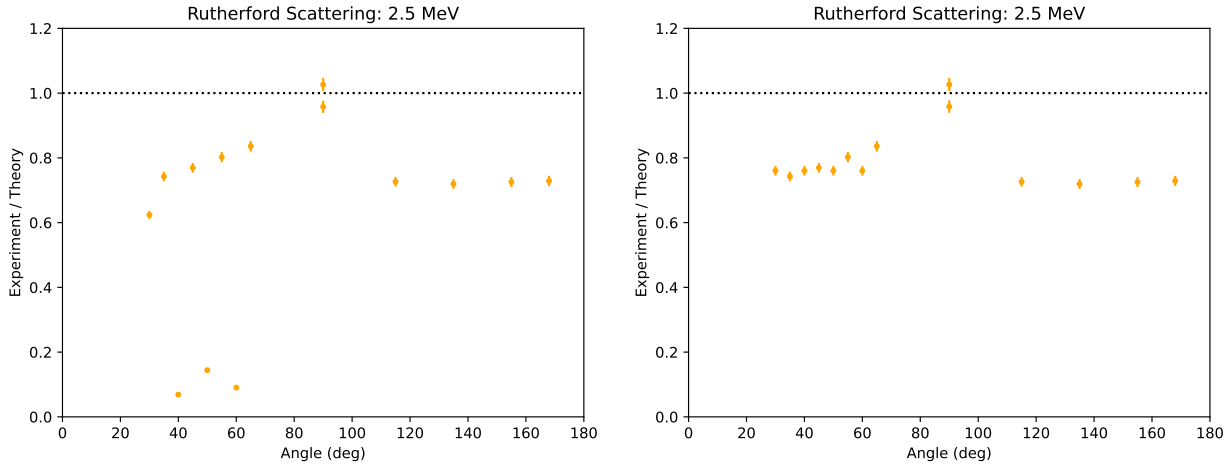
Absolute Angle	Original Ratio	Original Dead-time %	Estimated Ratio	Expected Dead-time %
30	0.62	10.29	0.76	26.39
40	0.07	7.61	0.76	91.64
50	0.14	5.41	0.76	82.05
60	0.09	1.84	0.76	88.33



(a) Original dead-time correction.

(b) Estimated dead-time correction.

FIG. 7: Angular distribution of differential cross-section for 2.5 MeV alpha beam.



(a) Original dead-time correction.

(b) Estimated dead-time correction.

FIG. 8: Experimental vs. theoretical differential cross-section ratios for 2.5 MeV alpha beam.

V. CONCLUSIONS

Overall, this experiment was successful at showing how the Rutherford cross-section is proportional to $1/sin^4(\theta/2)$. Even though there was a large systematic shift in the magnitude of the experimental data, the angular distribution's behavior matched the trend of the theoretical model. The other major source of error was assumed to be from the dead-time that was recorded. After calculating an expected dead-time to match the trend of the more consistent data points, the 30, 40, 50, and 60 degree data points were adjusted. Seeing as how there are still some discrepancies among the lower angle measurements after the dead-time correction, it is likely the dead-time was not the only error responsible for the poor low angle measurements. Even more curious is the disagreement between the two 90 degree measurements and the rest of the angles. The 90 degree measurements agree with the theoretical calculations within 2-3 σ, but the rest of the data remains about 25% lower. A more in-depth knowledge of the data acquisition electronics is needed to determine if this can be explained by faulty equipment, over-saturation of the detectors, or a simple partial obstruction of the detector face.

The experiment was also successful in determining the composition of the scattering target from the kinematics of scattering off different nuclei. The target's composition was found to be 79.1(1.0)% Gold, 11.60(61)% Silver, and 9.27(86)% Copper. This was close to the originally assumed 80% Gold, 10% Silver, and 10% Copper composition, which shows that Rutherford back-scattering can be very useful in materials analysis

VI. REFERENCES

[1] G. Jensen, P.M. King, and J. Roche, *PHYS6751: Nuclear Graduate Lab Lab manual v24.5 (Ohio University, 2024)*

[2] NIST, ASTAR, https://physics.nist.gov/PhysRefData/Star/Text/ASTAR.html